

Teaching a Small LLM Scholastic Voice

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Independent Research — github.com/pleyva2004/scholastic-llm

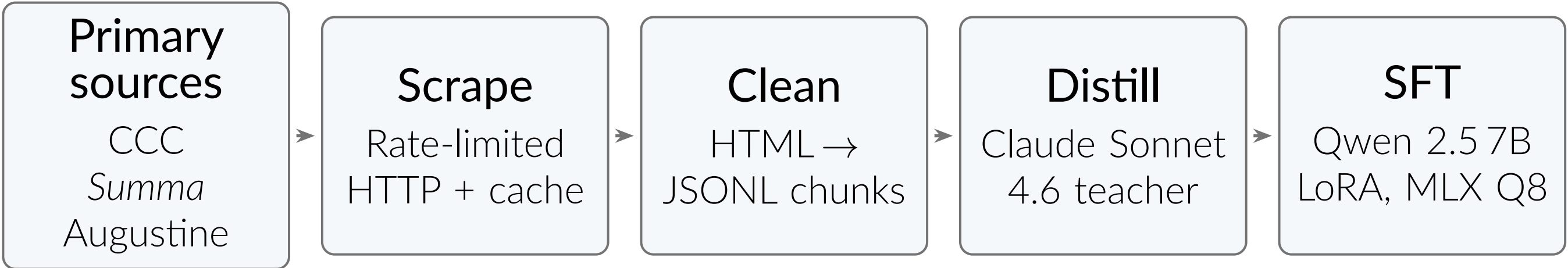
Motivation

Modern instruction-tuned LLMs default to flat, encyclopedic prose. Specialized historical writing has a distinct lexical and structural fingerprint: the Summa’s *quaestio* form, Augustine’s autobiographical address, the Catechism’s numbered paragraph citations.

Question. Can a small open-weights model (7B parameters) be moved into this scholastic register *cheaply*, on a single laptop, using only public-domain or fair-use source texts?

Answer. Yes. With teacher-distilled supervised fine-tuning on ~80 (question, scholastic-answer) pairs, a rubric measuring register, citation grounding, and argument structure rises from **19/120 to 68/120** (+258%) on 10 held-out philosophical prompts.

Pipeline



- Scrape CCC (vatican.va), Summa, Augustine (newadvent.org) with rate-limiting and on-disk cache.
- Clean into 342 structured JSONL chunks.
- Claude Sonnet 4.6 as *teacher*: per chunk, generate 2–3 (question, scholastic-answer) pairs with explicit register and CCC-citation instructions. 93 pairs, ≈\$0.50.
- Supervised fine-tune Qwen 2.5 7B-Instruct under MLX.

Training Setup

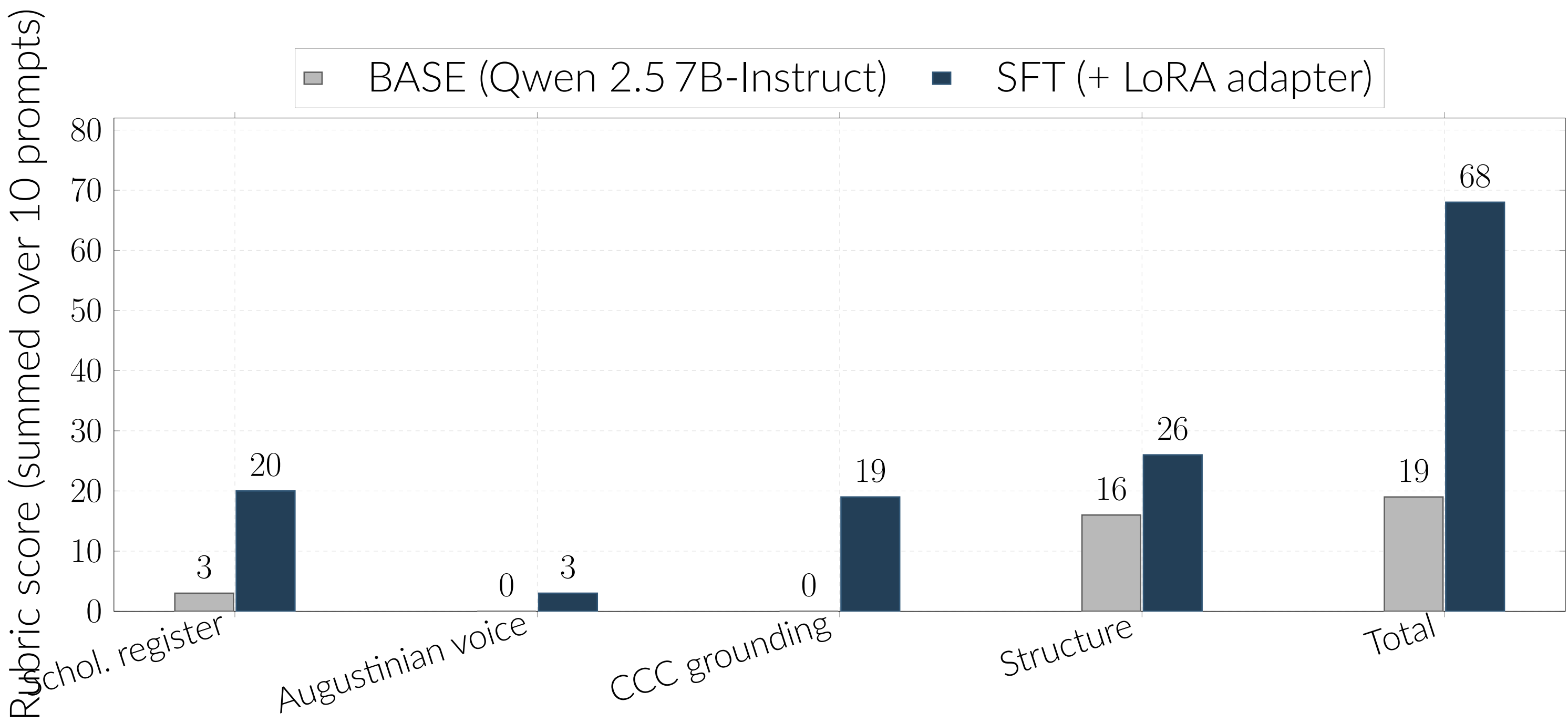
- Base: **Qwen2.5-7B-Instruct**, MLX 8-bit quantization
- LoRA on top 16 of 28 transformer layers
- AdamW, LR 10^{-5} , batch size 1, max seq 2048
- 200 iterations, 83 train + 10 valid
- Hardware: Apple M4 Pro, 48 GB unified memory
- Wall time: ≈ 7 minutes
- Peak resident memory: 12.7 GB
- Trainable parameters: 2.6M / 7.6B (0.034 %)

Eval prompts (held out)

- What is the nature of evil?
- Is faith reasonable?
- What is the relationship between grace and free will?
- How does the soul relate to the body?
- Why does God permit suffering?
- Can a human truly know God?
- What does it mean to love one’s neighbor?
- Is there such a thing as natural law?
- How do you reconcile divine foreknowledge with free will?
- What is the meaning of human life?

None of these appear in the 93 teacher-generated training pairs.

Results: rubric scores



Four-dimension rubric (0–3 each, total 0–12 per prompt) summed across 10 held-out philosophical questions ($N = 10$, max 120). Largest deltas: **scholastic register +17**, **CCC grounding +19**, **structure +10**.

Qualitative example

Q. How do you reconcile divine foreknowledge with free will?

BASE (Qwen 2.5 7B-Instruct): “The reconciliation of divine foreknowledge with free will is a classic philosophical and theological problem that has been debated for centuries. This issue arises from the apparent tension between the idea that a divine being has perfect knowledge of all future events...”

SFT (Qwen 2.5 7B + LoRA on 83 scholastic pairs): “This is perhaps the most ancient and most pressing question in all of metaphysics, and it is one that the Angelic Doctor addresses with great precision.

Objection 1. It would seem that divine foreknowledge destroys free will, for if God knows...”

Takeaway

Small open-weights models can be moved into a niche historical register cheaply when capacity is sufficient (7B not 1.5B) and quantization is gentle (Q8 not Q4). The bottleneck is not the optimization recipe but the *teacher signal*: a stronger model generating tightly-prompted training pairs.

Phase 2: scaling & voice rebalancing

Phase 2 scales to 377 training pairs (~4× Phase 1) with per-source register hints (Aquinas form for *Summa*/CCC chunks, Augustinian for *Confessions*/*City of God*), 800 SFT iterations. Best checkpoint: **iter 400** (mild overfitting after).

Augustinian gap closed: **aug** dim 3 → 7–13. The model now switches register per prompt rather than forcing every answer into Summa form.

Strict rubric undervalues this (penalizes Augustinian answers on **scholastic_register**). We add a **register_fluency** = **max(reg, aug)** metric and a *balanced total*; under it, Phase 2 reaches **68/90 vs Phase 1’s 66/90**.

Negative result: DPO saturation

The Phase 2 DPO chain (chosen = SFT-v2 output, rejected = base output) was inert. From iteration 1: val loss **0.000**, val accuracy **1.000**, chosen/rejected margin **35.4 nats**. The SFT policy already separated chosen from rejected with overwhelming margin, so DPO had no gradient signal.

Lesson. Within-model preference data is degenerate: when the policy generated chosen and the base generated rejected, the DPO objective is satisfied at initialization. Useful DPO signal requires preference pairs the policy has not already pre-learned to favor or disfavor.

Resources

- Repo:** github.com/pleyva2004/scholastic-llm
- Paper:** pleyva2004.github.io/scholastic-llm/main.pdf
- License:** code MIT; trained adapter not redistributed
- Research experiment, not theological authority.*



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